

Tsun-Yi Yang

shamangary@hotmail.com

<https://www.linkedin.com/in/tsunyyang/>

<https://github.com/shamangary>

<https://scholar.google.com/citations?user=WhISCE4AAAAJ>

PERSONAL SUMMARY

Tsun-Yi Yang is currently the founder and CEO of MIMI AI in the UK, a GenAI based physical data multi-model agent company. He is a researcher who focuses on spatial intelligence, synthetic data generation, vision perception (pose estimation, 3d world understanding) for 10yrs+ research experience. He has worked as a Meta/Facebook Research Scientist for 4 years, researching billion scale computer vision problems in Amazon, and he also worked at two startups as a research scientist (both got acquired). He led several projects with senior/junior researchers and interns. He has a Ph.D. from National Taiwan University, CSIE, with supervision from Dr. Yen-Yu Lin and Dr. Yung-Yu Chuang. He served the community as Program Committee (PC) in Association for the Advancement of Artificial Intelligence (AAAI) and was an award winning reviewer (outstanding reviewer) in several international conferences. He was a Ph.D. intern at a startup company Scape Technologies under the supervision of Dr. Vassileios Balntas, and the startup later was acquired by Facebook/Meta in 2020. His master is about quantum gate control in simulation. He published in top-tier conference papers with 1000+ citations including top conferences (CVPR, ECCV, ICCV ...) with more than 2000+ stars at github.

Skills: Pytorch, Huggingface, Python, Numpy, Pandas, Plotly, [Modal.com](https://modal.com) (large-scale deployment), AWS, GCP

HIGHLIGHT EXPERIENCE

+ [Large scale data curation and synthesis]

Tsun-Yi **led multiple senior scientists** in generating synthetic multi-model data generation for spatial intelligence <https://huggingface.co/datasets/projectaria/aria-synthetic-environments> which directly contributed to the pre-training and post-training vision language transformer from scratch and led to a top-conference paper in computer science.

<https://ai.meta.com/blog/scenescript-3d-scene-reconstruction-reality-labs-research/>

<https://eccv.ecva.net/virtual/2024/poster/200>

+ [CPU runnable physical library for robotics with blender and no other dependency]

Tsun-Yi personally built a blender based, cpu runnable on laptop, robotics simulation. Covering navigation, rigid-body physics, forces interaction. Different labeling modalities including masks, pointclouds, and more.

<https://github.com/mimiaigen/vibephysics>

+ [Regulated domain post-training]

Tsun-Yi **led cross domain teams including legal, product, and engineering** along with collaboration with Anthropic and AWS team. Post-training outperforming OpenAI o1-preview in legal domain with extensive analysis.

<https://robinai.com/news-and-resources/news/robin-ai-and-anthropic-claude-3-5-sonnet-are-top-performing-legal-ai-models>

EXPERIENCE

Founder and CEO, MIMI AI LTD

Mar 2025 – Present

Unstructured physical data Genai platform with multi-model agent and video gen world simulation

+ Scalable video-to-video api product. 20x faster than Nvidia-cosmos-transfer-2.5.

+ Multi-model agent with cross modality control (mesh, 3DGS) for scene composition and physical simulation library. <https://github.com/mimiaigen/vibephysics>

+ Customer applied our tech to syn-to-real with modern VLA controlling the robot truck.

https://www.linkedin.com/posts/bonsai-robotics_agricultureinnovation-computervision-ai-activity-7409703723696058368-gaeV/?utm_source=share&utm_medium=member_desktop&rcm=ACoAABQuet4BUoBxndKx-GvZKYMmE_P3sTuulY

Applied Scientist, Amazon, London

June 2025 - Oct 2025

- + Large-scale defect detection in Amazon warehouse with computer vision. Impacting several million Amazon packages everyday.
- + Multi-model research and application. For higher efficiency and robustness within the shipping process.
- + Low cost real-time keypoint based ROI segmentation to replace heavy transformer layer.

LLM Research Engineer (senior), Robin AI, London

Apr 2024 – Mar 2025

Legal domain LLM research

- + Leading junior engineer and legal expert with multi-model reasoning tasks with reinforcement learning. Tackling ambiguous visual to high level correspondence.
- + Leading junior engineer and legal expert in multi-model synthetic data quality and quantity including generation, evaluation, and training.
- + LLM fine-tune for legal co-pilot. Performance evaluation and weakness spotting.

Senior Research Scientist, Meta Reality Labs, London

Feb 2020 – Apr 2024

- + Leading research projects for competing with state-of-the-art research methods for AR/VR technologies
- + Improving 3d map building system with deep neural network for local and global feature learning (image matching, image retrieval). The improved features lead to lesser noise and end up with a more robust 3d map.
- + Leading/co-leading 5 interns ending up with 4 top-tier conference papers such as CVPR.

<https://psarlin.com/orienternet/>

https://openaccess.thecvf.com/content/CVPR2022/papers/Ng_NinjaDesc_Content-Concealing_Visual_Descriptor_s_via_Adversarial_Learning_CVPR_2022_paper.pdf

- + Diffusion generation model for large-scale 3d scene with deep-learning and transformer. Generating diverse indoor data such as non-manhattan layouts for better 3d network training.
- + Body motion reinforcement learning for natural walking with deep neural network in zero-shot (unseen) synthetic scene. Improving unnatural algorithm based trajectory generation for motion.
- + Serving as international conference reviewer for machine learning top conferences.

Computer vision researcher, Scape, London

Dec 2019 – Feb 2020

- + (This role quickly transferred to Meta research scientist due to acquisition of the startup.)
- + Implementation for start-up commercial proof-of-concept prototyping
- + Serving as international conference reviewer for machine learning top conferences

PhD research intern, Scape, London

Jan 2019 – Jul 2019

- + Serving as international conference reviewer for machine learning top conferences.
- + 2 papers published for deep neural network learning similarity search (image matching, image retrieval).
- + Collaboration team lead.

PhD research assistant, Academia Sinica, Taiwan

Oct 2013 – Dec 2018

- + Serving as international conference reviewer for machine learning top conferences.
- + 4 top-tier conference papers published.

Master of Applied Physics

Oct 2011 - Oct 2013

- + Simulated environment for quantum systems.

EDUCATION

National Taiwan University

2019

PhD in Computer Science

National Taiwan University

2013

MS in Applied Physics

SELECTED PUBLICATIONS/PROJECTS

+ SceneScript: Reconstructing Scenes With An Autoregressive Structured Language Model	ECCV2024
+ Aria Synthetic Environments Dataset, https://www.projectaria.com/datasets/ase/ ,	2023
+ OrienterNet: Visual Localization in 2D Public Maps with Neural Matching	CVPR2023
+ NinjaDesc: content-concealing visual descriptors via adversarial learning	CVPR2022
+ UR2KiD: Unifying Retrieval, Keypoint Detection, and Keypoint Description	ArXiv2020
+ FSA-Net: Learning Fine-Grained Structure Aggregation for Head Pose	CVPR2019
+ SSR-Net: A Compact Soft Stagewise Regression Network for Age Estimation	IJCAI2018
+ DeepCD: Learning Deep Complementary Descriptors for Patch Representations	ICCV2017

Awards, Services, Grants

2026, IEEE / CVF Computer Vision and Pattern Recognition Conference (CVPR), Reviewer
2023, IEEE / CVF Computer Vision and Pattern Recognition Conference (CVPR), Reviewer
2023, International Conference on Learning Representations (ICLR), Reviewer
2022, IEEE / CVF Computer Vision and Pattern Recognition Conference (CVPR), Reviewer
2022, IEEE / CVF Computer Vision and Pattern Recognition Conference (CVPR), Image Matching Workshop Reviewer
2021, IEEE / CVF Computer Vision and Pattern Recognition Conference (CVPR), Outstanding Reviewer
2021, IEEE / CVF Computer Vision and Pattern Recognition Conference (CVPR), Image Matching Workshop Reviewer
2021, IEEE / CVF International Conference on Computer Vision, Reviewer
2020, Association for the Advancement of Artificial Intelligence (AAAI), Program Committee (PC)
2020, Neural Information Processing Systems (Neurips), Reviewer
2019, IEEE / CVF Computer Vision and Pattern Recognition Conference (CVPR), Outstanding Reviewer
2019, IEEE / CVF International Conference on Computer Vision, Reviewer
2019, Association for the Advancement of Artificial Intelligence (AAAI), Reviewer
2018, Scholarship, Pan Wen Yuan Foundation